University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 2_____

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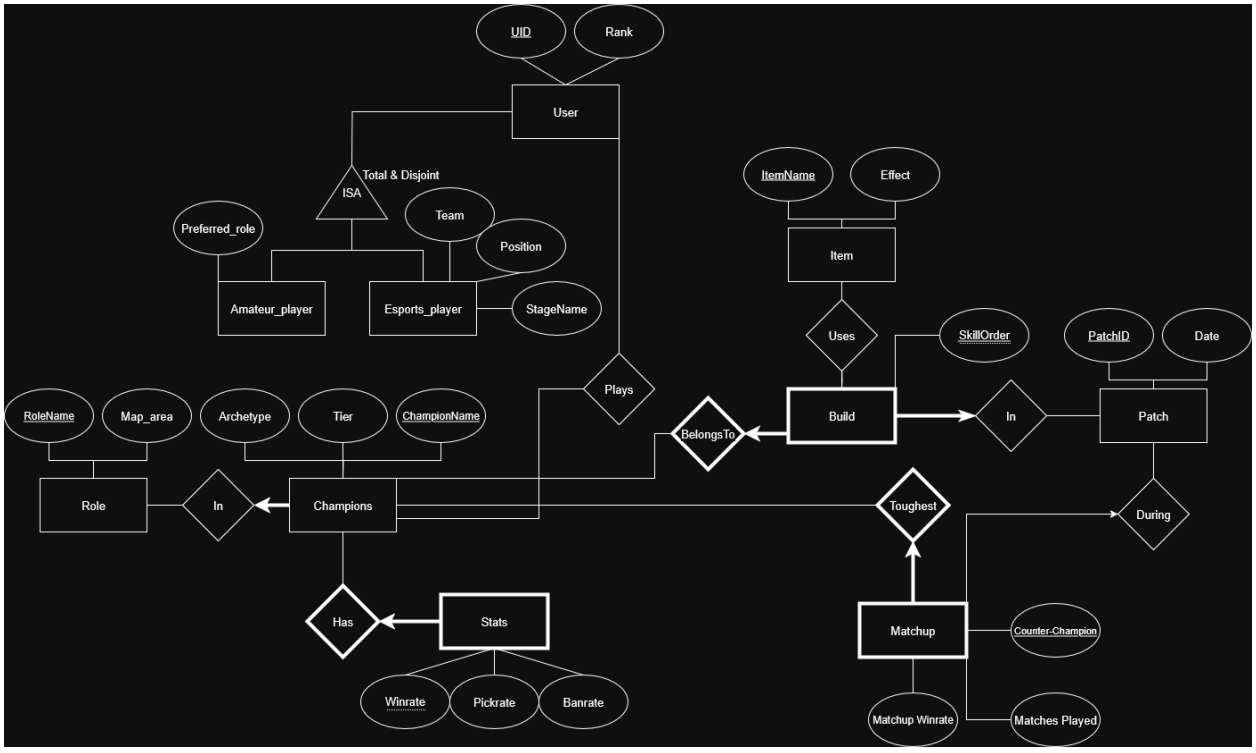
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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Summary:

Our project stores performance statistics for League of Legends champions, including win rate, pick rate and ban rate. It helps players make informed in-game decisions by offering insights into champion performance.



Changes of the ER diagram:

We used more specific attribute names across different entities to avoid confusion.
We have also changed stats and matchup to a weak entity to better reflect its relationship with champion.

Schema:

Amateur_player(UID: char, Rank: char, Preferred_role: char)

Esports_player(UID: char, Rank: char, Team: char, Position: char, StageName: char)

Plays(**UID**: char, **ChampionName**: char)

Champions(ChampionName: char, Tier: char, Archetype: char, **RoleName**: char)

Stats(Winrate: decimal, Pickrate: decimal, **ChampionName**: char, Banrate: decimal)

Role(RoleName: char, Map_area: char)

Matchup-during(**ChampionName**: char, Counter-Champion: char, Matchup Winrate: decimal, Matches Played: integer, **PatchID**: char)

Build(SkillOrder: char, **ChampionName**: char, **PatchID**: char)

Item(ItemName: char, Effect: char)

Build-Uses-Item(**SkillOrder**: char, **ChampionName**: char, **ItemName**: char)

Patch(PatchID: char, date: char)

FD:

Amateur_player:
UID -> Rank, UID -> Preferred_role

Esports_player:
UID -> Rank, Team, Position, stageName
Team -> Position, StageName

Role:
RoleName -> Map_area

Champions:
ChampionName -> Archetype, Tier, RoleName

Stats:
ChampionName, Winrate -> Pickrate, Banrate

Matchup:
ChampionName, Counter-Champion -> MatchupWinrate, MatchesPlayed, PatchID

Build:
SkillOrder, ChampionName -> PatchID

Item:
ItemName -> Effect

Patch:
PatchID -> date

Normalization (BCNF)

Team -> Position, Team is not a superkey, so Esports_player(UID, Rank, Team, Position, StageName) violates BCNF.
Decompose Esports_player(UID, Rank, Team, Position, StageName) to Esports_player1(UID, Rank, Team, stageName) and Esports_player2(Team, Position).

Team -> StageName, Team is not a superkey so Esports_player1(UID, Team, StageName) violates BCNF.
Decompose Esports_player1(UID, Rank, Team, stageName) to Esports_player1(UID, Rank, Team) and Esports_player3(Team, StageName).

Result: Esports_player1(<u>UID</u>, Rank, Team), Esports_player2(<u>Team</u>, Position), Esports_player3(<u>Team</u>, StageName)

SQL DDL statements

```
CREATE TABLE Amateur_player (
   UID            INTEGER PRIMARY KEY,
   Rank           VARCHAR(20),
   Preferred_role  VARCHAR(10),
);


CREATE TABLE Esports_player1 (
   UID       INTEGER PRIMARY KEY,
   Rank      VARCHAR(20),
   Team      VARCHAR(30),
);

CREATE TABLE Esports_player2 (
   Team        VARCHAR(30) PRIMARY KEY,
   Position    VARCHAR(20),
);


CREATE TABLE Esports_player3 (
   Team            VARCHAR(30) PRIMARY KEY,
   StageName    VARCHAR(20),
);

CREATE TABLE Champions (
   ChampionName       VARCHAR(30) PRIMARY KEY,
   Archetype              VARCHAR(30),
   Tier                       VARCHAR(10),
   RoleName              VARCHAR(20),
   FOREIGN KEY (RoleName) REFERENCES Role ON DELETE CASCADE
);




CREATE TABLE Stats (
   ChampionName  VARCHAR(50),
```

```
    Winrate       DECIMAL(5,2),
    Pickrate      DECIMAL(5,2),
    Banrate       DECIMAL(5,2),
    PRIMARY KEY (ChampionName , Winrate),
    FOREIGN KEY (ChampionName) REFERENCES Champions ON DELETE CASCADE
);

CREATE TABLE Role (
    RoleName      VARCHAR(20) PRIMARY KEY,
    Map_area      VARCHAR(10)
);

CREATE TABLE Patch (
    PatchID  DECIMAL(4,2) PRIMARY KEY,
    date       VARCHAR(10)
);

CREATE TABLE Item (
    Name    VARCHAR(50) PRIMARY KEY,
    Effect    VARCHAR(200)
);

CREATE TABLE Build (
    ChampionName  VARCHAR(50),
    SkillOrder          CHAR(3),
    PatchID             VARCHAR(10) NOT NULL,
    PRIMARY KEY (ChampionName, SkillOrder),
    FOREIGN KEY (ChampionName) REFERENCES Champions ON DELETE CASCADE,
    FOREIGN KEY (PatchID) REFERENCES Patch ON DELETE CASCADE
);

CREATE TABLE BuildUsesItem (
    ChampionName  VARCHAR(50),
    SkillOrder          VARCHAR(20),
    ItemName          VARCHAR(50),
    PRIMARY KEY (ChampionName, SkillOrder, ItemName),
    FOREIGN KEY (ChampionName, SkillOrder) REFERENCES Build ON DELETE CASCADE,
    FOREIGN KEY (ItemName) REFERENCES Item ON DELETE CASCADE
);



CREATE TABLE Matchup (
    ChampionName        VARCHAR(50),
```

```
    Counter-Champion   VARCHAR(50),
    PatchID            VARCHAR(10),
    MatchupWinrate     DECIMAL(5,2),
    MatchesPlayed      INTEGER,
    PRIMARY KEY (ChampionName, Counter-ChampionName),
    FOREIGN KEY (ChampionName) REFERENCES Champions ON DELETE CASCADE,
    FOREIGN KEY (PatchID) REFERENCES Patch ON DELETE CASCADE
);
```

Populate tables:

```
INSERT INTO Amateur_player (UID, Rank, Preferred_role) VALUES (101, 'Bronze', 'Mid');
INSERT INTO Amateur_player (UID, Rank, Preferred_role) VALUES (102,'Silver',  'Top');
INSERT INTO Amateur_player (UID, Rank, Preferred_role) VALUES (103, 'Gold', 'Support');
INSERT INTO Amateur_player (UID, Rank, Preferred_role) VALUES (104, 'Platinum', 'Jungle');
INSERT INTO Amateur_player (UID, Rank, Preferred_role) VALUES (105, 'Diamond', 'ADC');

INSERT INTO Esports_player1 (UID, Rank, Team) VALUES (201, 'Challenger', 'GenG');
INSERT INTO Esports_player1 (UID, Rank, Team) VALUES (202,'Challenger', 'T1');
INSERT INTO Esports_player1 (UID, Rank, Team) VALUES (203, 'Challenger', 'HLE');
INSERT INTO Esports_player1 (UID, Rank, Team) VALUES (204, 'Challenger', 'AL');
INSERT INTO Esports_player1 (UID, Rank, Team) VALUES (205, 'Challenger', 'BLG');

INSERT INTO Esports_player2 (Team, Position) VALUES ('GenG', 'Top');
INSERT INTO Esports_player2 (Team, Position) VALUES ('T1', 'Jungle');
INSERT INTO Esports_player2 (Team, Position) VALUES ('HLE', 'Mid');
INSERT INTO Esports_player2 (Team, Position) VALUES ('AL', 'ADC');
INSERT INTO Esports_player2 (Team, Position) VALUES ('BLG', 'Support');

INSERT INTO Esports_player3 (Team, StageName) VALUES ('GenG', 'Kiin');
INSERT INTO Esports_player3 (Team, StageName) VALUES ('T1', 'Oner');
INSERT INTO Esports_player3 (Team, StageName) VALUES ('HLE', 'Zeka');
INSERT INTO Esports_player3 (Team, StageName) VALUES ('AL', 'Hope');
INSERT INTO Esports_player3 (Team, StageName) VALUES ('BLG', 'ON');

INSERT INTO Champions(ChampionName, Tier, Archetype, RoleName) VALUES ('Ambessa', 'OP',
'Fighter', 'Top')
INSERT INTO Champions(ChampionName, Tier, Archetype, RoleName) VALUES ('Pantheon', '1',
'Fighter', 'Jungle')
INSERT INTO Champions(ChampionName, Tier, Archetype, RoleName) VALUES ('Azir', '1', 'Mage',
'Mid')
```

INSERT INTO Champions(ChampionName, Tier, Archetype, RoleName) VALUES (‘Ezreal’, ‘2’, ‘Marksman’, ‘ADC’)
INSERT INTO Champions(ChampionName, Tier, Archetype, RoleName) VALUES (‘Alistar’, ‘OP’, ‘Tank’, ‘Support’)

INSERT INTO Stats(ChampionName, Winrate, Pickrate, Banrate) VALUES (‘Alistar’, ‘54.69%’, ‘19.55%’, ‘4.20%’)
INSERT INTO Stats(ChampionName, Winrate, Pickrate, Banrate) VALUES (‘Ezreal’, ‘47.14%’, ‘11.72%’, ‘2.50%’)
INSERT INTO Stats(ChampionName, Winrate, Pickrate, Banrate) VALUES (‘Azir’, ‘52.23%’, ‘10.44%’, ‘2.38%’)
INSERT INTO Stats(ChampionName, Winrate, Pickrate, Banrate) VALUES (‘Pantheon’, ‘52.04%’, ‘16.51%’, ‘19.58%’)
INSERT INTO Stats(ChampionName, Winrate, Pickrate, Banrate) VALUES (‘Ambessa’, ‘50.94%’, ‘13.67%’, ‘5.88%’)

INSERT INTO Matchup(ChampionName, CounterChampion, PatchID, MatchupWinrate, MatchesPlayed) VALUES (‘‘Ambessa’’, ‘Gwen’, ‘46.15%’, ‘39’)
INSERT INTO Matchup(ChampionName, CounterChampion, PatchID, MatchupWinrate, MatchesPlayed) VALUES (‘Pantheon’,Trundle, ‘51.35%’, ‘37’)
INSERT INTO Matchup(ChampionName, CounterChampion, PatchID, MatchupWinrate, MatchesPlayed) VALUES (‘Azir’,Taliyah, ‘42.55%’, ‘47’)
INSERT INTO Matchup(ChampionName, CounterChampion, PatchID, MatchupWinrate, MatchesPlayed) VALUES (‘Kai’Sa’,Lucian, ‘43.33%’, ‘30’)
INSERT INTO Matchup(ChampionName, CounterChampion, PatchID, MatchupWinrate, MatchesPlayed) VALUES (‘Alistar’, ‘Braum’, ‘39.39%’, ‘33’)

INSERT INTO Role (RoleName, Map_area) VALUES (‘Top’, ‘Top Lane’)
INSERT INTO Role (RoleName, Map_area) VALUES (‘Jungle’, ‘The Jungle’)
INSERT INTO Role (RoleName, Map_area) VALUES (‘Mid’, ‘Mid Lane’)
INSERT INTO Role (RoleName, Map_area) VALUES (‘ADC’, ‘Bot Lane’)
INSERT INTO Role (RoleName, Map_area) VALUES (‘Support’, ‘Bot Lane’)

INSERT INTO Patch (PatchID, date) VALUES (‘25.14’, ‘2025-07-15’)
INSERT INTO Patch (PatchID, date) VALUES (‘25.13’, ‘2025-06-24’)
INSERT INTO Patch (PatchID, date) VALUES (‘25.12’, ‘2025-06-10’)
INSERT INTO Patch (PatchID, date) VALUES (‘25.11’, ‘2025-05-27’)
INSERT INTO Patch (PatchID, date) VALUES (‘25.10’, ‘2025-05-13’)

INSERT INTO Item (Name, Effect) VALUES (‘Infinity Edge’, ‘Massively enhances critical strikes.’);
INSERT INTO Item (Name, Effect) VALUES (‘Luden"s Companion’, ‘Grants Ability Power, Mana, and burst damage.’);
INSERT INTO Item (Name, Effect) VALUES (‘Heartsteel’, ‘Grants stacking Health and HP scaling damage.’);

INSERT INTO Item (Name, Effect) VALUES (‘Redemption’, ‘ACTIVE: Heals allies and damages enemies in a circle.’);
INSERT INTO Item (Name, Effect) VALUES (‘Zhonya’s Hourglass’, ‘ACTIVE: Become invincible for the duration but unable to take action.’);
INSERT INTO Item (Name, Effect) VALUES (‘Eclipse’, ‘Grants Attack damage and a shield.’);

INSERT INTO Build (ChampionName, SkillOrder, PatchID) VALUES (‘Ambessa’, ‘QEW’, ‘15.14’);
INSERT INTO Build (ChampionName, SkillOrder, PatchID) VALUES (‘Alistar’, ‘QWE’, ‘15.13’);
INSERT INTO Build (ChampionName, SkillOrder, PatchID) VALUES (‘Ezreal’, ‘QEW’, ‘15.12’);
INSERT INTO Build (ChampionName, SkillOrder, PatchID) VALUES (‘Azir’, ‘WQE’, ‘15.11’);
INSERT INTO Build (ChampionName, SkillOrder, PatchID) VALUES (‘Pantheon’, ‘QWE’, ‘15.10’);

INSERT INTO BuildUsesItem (ChampionName, SkillOrder, PatchID, ItemName) VALUES (‘Ambessa’, ‘QEW’, ‘15.14’,‘Heartsteel’);
INSERT INTO BuildUsesItem (ChampionName, SkillOrder, PatchID, ItemName) VALUES (‘Alistar’, ‘QWE’, ‘15.13’, ‘Redemption’);
INSERT INTO BuildUsesItem (ChampionName, SkillOrder, PatchID, ItemName) VALUES (‘Ezreal’, ‘QEW’, ‘15.12’, ‘Eclipse’);
INSERT INTO BuildUsesItem (ChampionName, SkillOrder, PatchID, ItemName) VALUES (‘Azir’, ‘WQE’, ‘15.11’, ‘Zhonya’s Hourglass’);
INSERT INTO BuildUsesItem (ChampionName, SkillOrder, PatchID, ItemName) VALUES (‘Pantheon’, ‘QWE’, ‘15.10’, ‘Infinity Edge’);